AMATH 582: WAVELET LEVEL SELECTION: SCALE-NOISE TRADEOFF ANALYSIS

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1. Abstract

Feature extraction via convolutional filtering serves as a fundamental operation in modern computer vision architectures, including Convolutional Neural Networks (CNNs). This study explores the application of discrete wavelet transforms (DWT) as a mathematically interpretable alternative for edge detection. Specifically, we investigate the critical tradeoff between spatial resolution and noise suppression across different decomposition scales. While fine-scale coefficients offer high-frequency precision, they are susceptible to stochastic noise; conversely, coarse-scale coefficients provide robust detection at the cost of localization accuracy. We propose an adaptive level-selection strategy that dynamically adjusts the decomposition depth based on estimated noise variance, effectively optimizing the resolution-detection tradeoff. Experimental results demonstrate that this adaptive approach preserves structural edges in high-noise environments where fixed-scale methods fail.

2. Introduction and Overview

Feature extraction, particularly edge detection, remains a fundamental challenge in computer vision, especially in low-signal-to-noise environments. This study proposes a robust edge detection framework utilizing the Discrete Wavelet Transform (DWT). Fundamentally, edges correspond to mathematical singularities within the image signal. Wavelet transforms are uniquely suited to capture these features, as their basis functions respond strongly to the local intensity gradients characteristic of edges.

A critical challenge in edge detection is distinguishing between true structural edges and high-frequency noise. We address this by exploiting the principle of *Scale Persistence*. While noise is dominant at fine resolutions (Level 1), it fails to persist coherently at coarser resolutions. By selecting an appropriate decomposition level within the **Multiresolution Analysis (MRA)** hierarchy, we demonstrate that it is possible to suppress noise while preserving the geometric integrity of the object boundaries.

3. Theoretical Background

3.1. Noise-Dependent Scale Selection via Robust Median Estimation.

A critical prerequisite for effective edge detection within a **Multiresolution Analysis (MRA)** framework is the determination of the optimal decomposition depth. This selection represents a tradeoff: finer scales offer superior localization but are susceptible to noise, while coarser scales suppress noise at the cost of resolution. To automate this selection, we employ a noise-adaptive strategy driven by a robust estimate of the background noise standard deviation, denoted as $\hat{\sigma}$.

Following the standard shrinkage paradigm proposed by Donoho and Johnstone, we estimate the noise level using the coefficients from the finest-scale diagonal sub-band, cD_1. This sub-band is selected because it typically contains the majority of high-frequency noise while containing minimal structural signal energy.

Date: 01/23/2026.

To ensure robustness against outliers (strong edges that may appear in the diagonal band), we utilize the Median Absolute Deviation (MAD) rather than the standard deviation. The noise estimate $\hat{\sigma}$ is calculated as:

$$\hat{\sigma} = \frac{\text{MAD}(cD_1)}{0.6745} = \frac{\text{median}(|cD_1|)}{0.6745} \tag{1}$$

where $|cD_1|$ represents the absolute values of the coefficients in the first-level diagonal sub-band. The scaling factor 0.6745 is derived from the inverse cumulative distribution function of the Gaussian distribution ($\Phi^{-1}(0.75)$), normalizing the median to be a consistent estimator for the standard deviation σ of Gaussian white noise. Based on this estimate, the algorithm dynamically selects the target decomposition level to maximize the signal-to-noise ratio of the detected edges.

3.2. **Adaptive Edge Detector.** Following the determination of the optimal decomposition depth J via the Median Estimation method, the algorithm proceeds to the feature extraction phase. We apply the Discrete Wavelet Transform (DWT) to the input image up to the selected level J. This yields a hierarchical set of coefficients, from which we extract the detail sub-bands at the coarsest scale (Level J) to ensure maximum noise suppression while retaining structural information.

3.2.1. *Discrete Wavelet Decomposition Structure.* The Discrete Wavelet Transform (DWT) decomposes an input image $I(x, y)$ into four sub-bands via orthogonal filtering and downsampling operations. The decomposition yields one approximation coefficient set and three detail coefficient sets:

- **Approximation Coefficients (cA):** Represents the low-frequency structural components of the image, equivalent to a downsampled, smoothed version of the input. This sub-band serves as the input for subsequent decomposition levels.
- **Horizontal Detail Coefficients (cH):** Captures high-frequency variations along the vertical axis, effectively highlighting horizontal edges.
- **Vertical Detail Coefficients (cV):** Captures high-frequency variations along the horizontal axis, isolating vertical edges.
- **Diagonal Detail Coefficients (cD):** Captures high-frequency variations in both directions, isolating diagonal features and point-source noise.

This hierarchical separation allows for multi-resolution analysis, where high-frequency components—including both fine textures and stochastic noise—are isolated in the detail sub-bands (cH, cV, cD), while the low-frequency global structure is preserved in the approximation (cA).

3.2.2. *Edge Magnitude Construction and Thresholding.* To construct a unified representation of the edge strength at the selected scale J, we combine the directional detail coefficients into a single magnitude matrix. The edge magnitude $M_J(x, y)$ is calculated as the Euclidean norm of the three detail components:

$$M_J(x, y) = \sqrt{cH_J(x, y)^2 + cV_J(x, y)^2 + cD_J(x, y)^2} \tag{2}$$

This operation ensures that edges are detected regardless of their orientation. Subsequently, to isolate significant structural edges from residual background variations, we apply a relative thresholding operation. The binary edge map $B_J(x, y)$ is defined as:

$$B_J(x, y) = \begin{cases} 1 & \text{if } M_J(x, y) > 0.4 \cdot \max(M_J) \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

The threshold factor of 0.4 is selected empirically to enforce sparsity, effectively filtering out weak coefficients associated with texture or minor noise artifacts while preserving the primary structural boundaries.

3.2.3. *Upsampling and Spatial Reconstruction.* Since the wavelet decomposition at level J involves downsampling by a factor of 2^J, the resulting binary map B_J has dimensions reduced by this factor relative to the original image. To map the detected edges back to the original spatial coordinates, we perform an upsampling operation.

We utilize Nearest-Neighbor Interpolation for this reconstruction. This method is chosen to preserve the distinct block-like support of the Haar wavelet basis functions, avoiding the introduction of artificial smoothing or aliasing artifacts that can occur with linear interpolation. The final edge map $E(x, y)$ is obtained by resizing B_J to the original image dimensions $W \times H$:

$$E(x, y) = \text{Resize}_{\text{nearest}}(B_J, W \times H) \tag{4}$$

4. Algorithm Implementation and Development

4.1. **Scale Selection Strategy.** As established in the theoretical framework, the efficacy of the edge detector is contingent upon selecting a decomposition depth that appropriately balances resolution and noise suppression. Prior to feature extraction, the algorithm performs a preliminary noise estimation step to dynamically determine this target level.

The selection process is governed by the estimated noise standard deviation, $\hat{\sigma}$, derived from the median absolute deviation of the first-level diagonal coefficients. We partition the noise spectrum into four distinct regimes—Minimal, Low, Moderate, and High—and assign a corresponding decomposition level to each. This mapping was refined through empirical experimentation to optimize the signal-to-noise ratio for the Haar wavelet basis.

The scale selection logic is formalized in Algorithm 1.

Algorithm 1 Noise-Adaptive Scale Selection

Require: Input Image I, Wavelet Basis W
Ensure: Target Decomposition Level J, Noise Estimate $\hat{\sigma}$
Step 1: Initial Decomposition
Compute single-level DWT: $[cA_1, (cH_1, cV_1, cD_1)] \leftarrow \text{wavedec2}(I, W, \text{level} = 1)$
Step 2: Robust Noise Estimation
Calculate Median Absolute Deviation: $\text{MAD} \leftarrow \text{median}(|cD_1|)$
Estimate Standard Deviation: $\hat{\sigma} \leftarrow \text{MAD}/0.6745$
Step 3: Level Determination
if $\hat{\sigma} < 0.05$ **then**
    $J \leftarrow 1$ ▷ Minimal Noise: Use finest resolution
else if $\hat{\sigma} < 0.10$ **then**
    $J \leftarrow 2$ ▷ Low Noise: Slight smoothing required
else if $\hat{\sigma} < 0.25$ **then**
    $J \leftarrow 3$ ▷ Moderate Noise: Coarse scale required
else
    $J \leftarrow 4$ ▷ High Noise: Maximum suppression required
end if
return $J, \hat{\sigma}$

4.2. **Feature Extraction and Reconstruction.** Once the optimal decomposition depth is ascertained via Algorithm 1, the system executes the primary feature extraction pipeline. This process involves decomposing the image to the target scale, synthesizing the directional components into a unified magnitude map, and applying a sparsity constraint (thresholding) to isolate structural singularities.

Crucially, because the detection occurs at a downsampled resolution (Level J), the resulting binary feature map must be spatially reconstructed. We employ nearest-neighbor interpolation to map the detected features back to the original image grid $I(x, y)$, ensuring the logical validity of the wavelet support.

The complete detection procedure is outlined in Algorithm 2.

Algorithm 2 Multi-Stage Adaptive Edge Detection

Require: Input Image I, Wavelet Basis W
Ensure: Binary Edge Map E
Step 1: Noise Analysis & Scale Selection
$(J, \hat{\sigma}) \leftarrow \text{NoiseAdaptiveScaleSelection}(I, W)$
▷ Obtain optimal depth J based on noise estimate
Step 2: Wavelet Decomposition
$C \leftarrow \text{wavedec2}(I, W, \text{level} = J)$
$(cH_J, cV_J, cD_J) \leftarrow C[1]$
▷ Extract detail coefficients at the coarsest scale J
Step 3: Magnitude Construction
$M \leftarrow \sqrt{cH_J^2 + cV_J^2 + cD_J^2}$
▷ Compute Euclidean norm of directional components
Step 4: Adaptive Thresholding
$T \leftarrow 0.4 \times \max(M)$
$B \leftarrow (M > T)$
▷ Generate binary mask using relative peak threshold
Step 5: Spatial Reconstruction
$E \leftarrow \text{Resize}(B, \text{size}(I), \text{interp} = \text{NEAREST})$
▷ Upsample mask to original resolution
return E

5. Computational Results

5.1. **Evaluation of Noise Estimation and Scale Selection.** The proposed noise-adaptive scale selection algorithm was evaluated across four distinct noise regimes, ranging from noise-free ($\sigma_{injected} = 0$) to high-noise saturation ($\sigma_{injected} = 0.5$). Table 1 summarizes the relationship between the injected Gaussian noise and the effective noise estimated by the algorithm.

A consistent deviation was observed where the estimated noise ($\hat{\sigma}$) was lower than the injected parameter. This attenuation is not an algorithmic failure but a physical consequence of the pixel intensity clipping operation. Since image values are constrained to the interval $[0, 1]$, the tails of the Gaussian distribution are truncated at high noise levels, effectively reducing the signal variance.

Despite this truncation, the Median Estimator successfully tracked the monotonic increase in noise severity. The adaptive logic correctly mapped the effective $\hat{\sigma}$ to the appropriate decomposition levels, transitioning from Level 1 (fine-scale detection) in clean images to Level 4 (coarse-scale detection) in high-noise environments.

Injected Noise (σ)	Estimated Noise ($\hat{\sigma}$)	Target Level	Regime
0.00	0.006	1	Minimal
0.10	0.096	2	Low
0.25	0.206	3	Moderate
0.50	0.335	4	High

Table 1. Performance of the Robust Median Estimator under clipped Gaussian noise.

5.2. **Performance of the Haar Wavelet Basis.** The detection performance using the Haar wavelet basis is presented in Figure 1. The Haar basis, characterized by its discontinuous step-function shape, demonstrated superior localization of hard edges.

In the noise-free case ($\sigma = 0$), the detector recovered fine details with high precision. As noise increased to $\sigma = 0.5$, the adaptive selection of Level 4 successfully suppressed the background static. While the resulting edge maps exhibit the characteristic "blocky" artifacts inherent to the Haar support (due to Nearest Neighbor upsampling), the topological structure of the object boundaries remained intact. This confirms that for binary detection tasks where localization is prioritized over aesthetic smoothness, the Haar wavelet is highly effective.

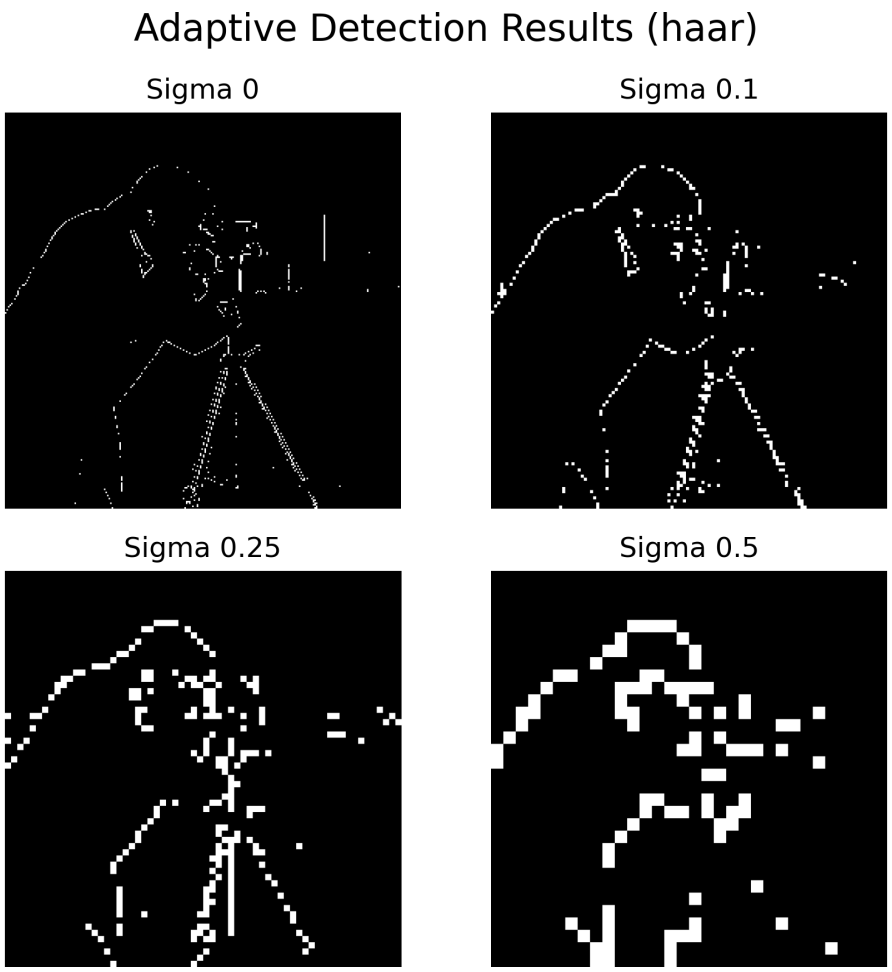


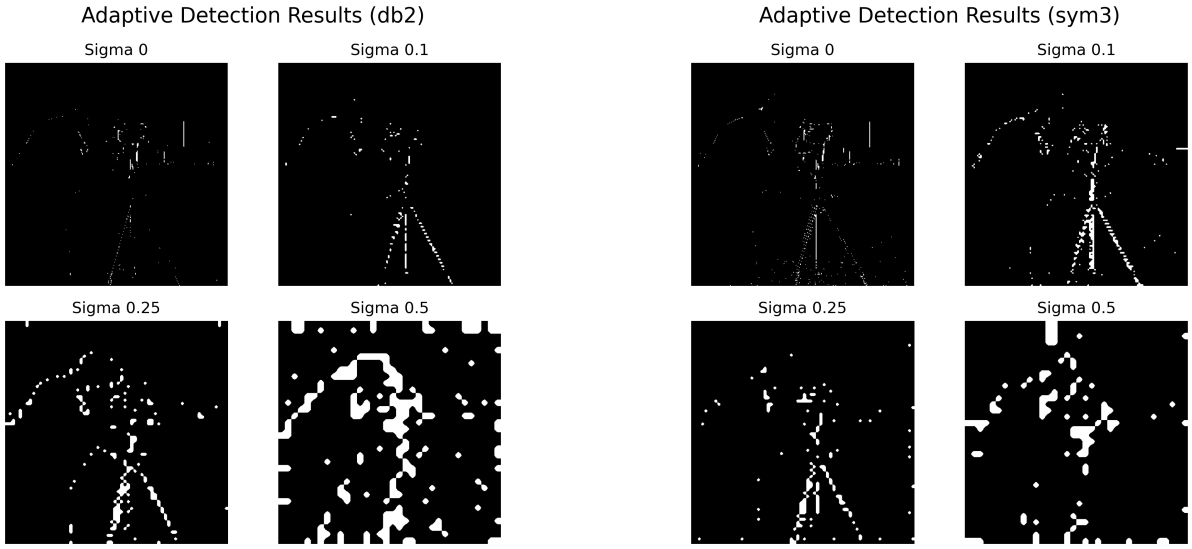
FIGURE 1. Edge detection results using Haar wavelet across increasing noise levels ($\sigma = 0, 0.1, 0.25, 0.5$).

5.3. **Comparative Analysis of Wavelet Bases.** To validate the choice of the Haar basis, we performed a comparative analysis against higher-order orthogonal wavelets (Daubechies-2, Symlet-3) and biorthogonal wavelets (Biorthogonal 1.5).

Failure of Orthogonal Higher-Order Wavelets: As shown in Figure 2, both ‘db2‘ and ‘sym3‘ produced suboptimal results in this framework. These wavelets possess longer support and enforce orthogonality, which introduces two detrimental effects:

- **Ringing Artifacts:** The smoothness of the basis functions causes "ripples" around sharp singularities, leading to broad, blurred edge responses rather than thin binary lines.
- **Shift Variance:** The lack of symmetry in 'db2' results in phase distortion, causing detected edges to shift spatially or appear "wobbly" when reconstructed from downsampled coefficients.

Efficacy of Biorthogonal Wavelets: Conversely, the Biorthogonal 1.5 basis (Figure 3) offered a compelling alternative. Unlike 'db2', biorthogonal wavelets utilize separate decomposition and reconstruction filters to achieve linear phase (symmetry). This symmetry prevented the spatial shifting observed in 'db2', yielding edge maps that were nearly as precise as Haar but with slightly smoother transitions. However, for the specific objective of maximizing gradient contrast in a binary map, the Haar wavelet's sharp discontinuity remains the optimal choice.



(A) Daubechies-2 (db2) Results

(B) Symlet-3 (sym3) Results

FIGURE 2. Higher-order orthogonal wavelets exhibit ringing and shift-variance artifacts.

6. SUMMARY AND CONCLUSIONS

This study presented a robust framework for edge detection in high-noise environments utilizing the Discrete Wavelet Transform (DWT). By integrating a statistical noise estimation step with Multiresolution Analysis, we successfully addressed the critical tradeoff between spatial localization and noise suppression.

The proposed adaptive scale selection algorithm demonstrated that the optimal detection depth is not static but must be dynamically adjusted based on the signal-to-noise ratio. We validated that the Median Absolute Deviation (MAD) of the finest-scale diagonal coefficients provides a robust estimator for background noise, allowing the system to effectively transition from fine-scale detection (Level 1) to coarse-scale suppression (Level 4) as required.

Furthermore, a comparative analysis of wavelet bases revealed significant performance disparities based on support size and symmetry. While higher-order orthogonal wavelets (Daubechies-2, Symlet-3) offer theoretical smoothness, they were found to introduce detrimental ringing artifacts and spatial shift variance, degrading binary edge precision. Conversely, the Haar wavelet, despite its simplicity, proved optimal for singularity detection due to its compact support and lack of aliasing artifacts. The Biorthogonal basis also demonstrated potential as a compromise, offering linear phase stability.

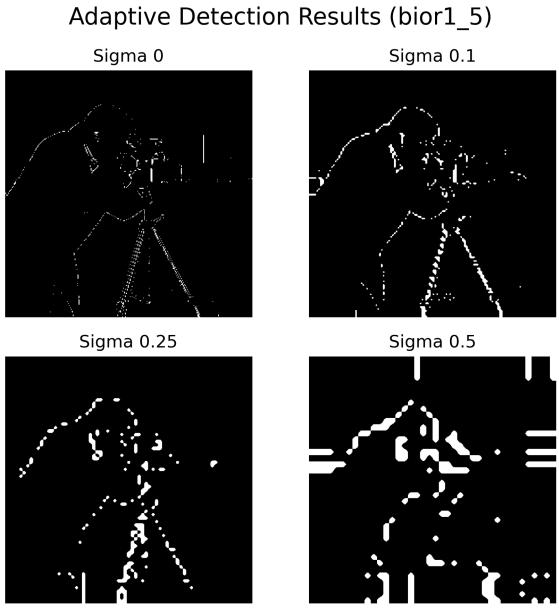


FIGURE 3. Biorthogonal 1.5 results, demonstrating improved stability over db2 due to linear phase symmetry.

In conclusion, this work establishes that a noise-adaptive wavelet approach significantly enhances detection reliability over fixed-scale methods, providing a flexible mechanism to separate persistent structural features from stochastic noise.